

\_\_\_\_\_

Page 10 of 10

Page 10 of 10

Page 10 of 10

Page 10 of 10

□ □ □ □ □

\_\_\_\_\_

KK3  
M3X8/D7985B

KK4  
M3X8/D7985B

KK5  
M3X8/D7985B

KK6  
M3X8/D7985B



KK7  
SK96/84SA



KK1  
PCB-MB-01

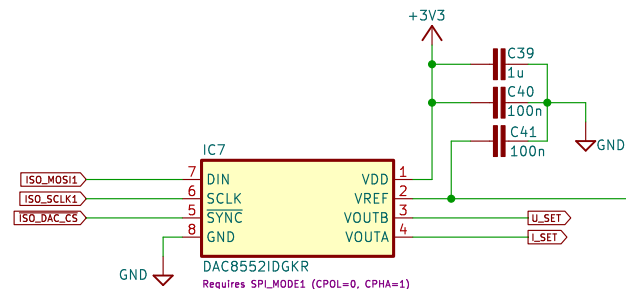
KK2  
PCB-MB-01



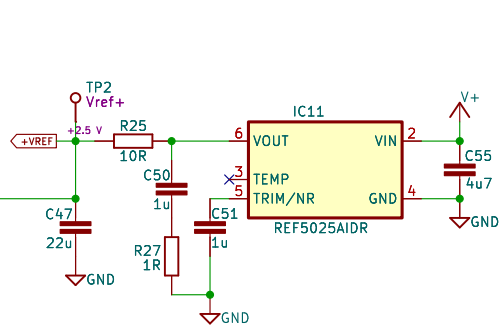
Id: 1/8



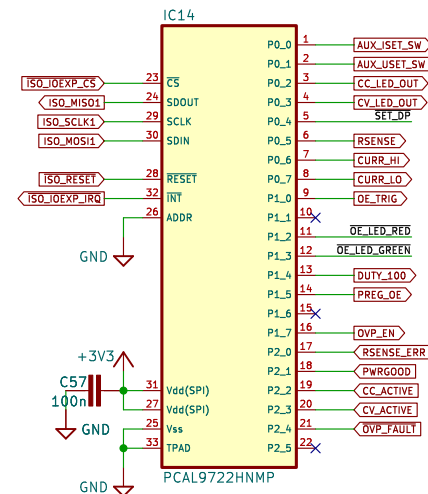
# 16-bit DAC (set voltage and current)



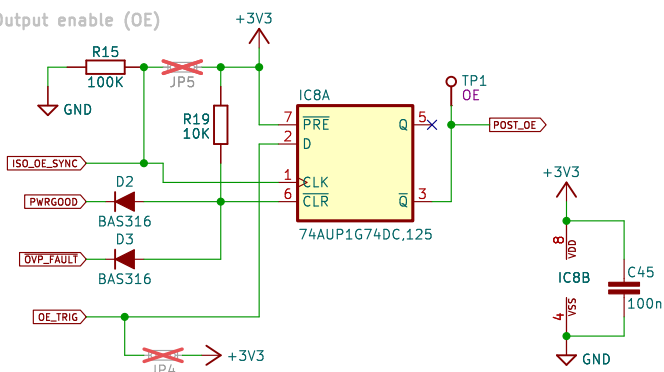
# Voltage reference



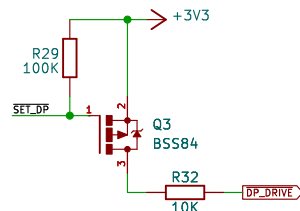
# SPI 22-bit I/O Expander



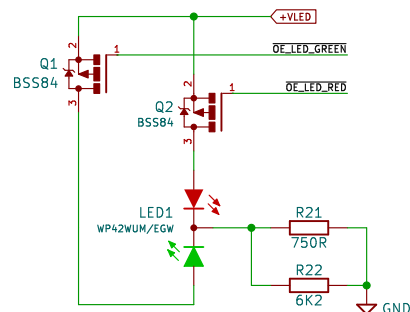
# Output enable (OE)



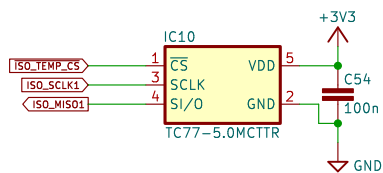
# Down-programmer



# OE indicator



# SPI Temperature sensor



<https://www.envox.eu>

Power module with SMPS pre-regulator, remote sense & programming  
Licensed under the TAPR Open Hardware License ([www.tapr.org/OHL](http://www.tapr.org/OHL))

**Envox d.o.o.**

Sheet: /DAC, OE, Vref, IOEXP, Temp/

File: DAC\_Vref\_IOEXP\_Temp.kicad\_sch

**Title: EEZ DIB DCP405+ power module**

Size: A4

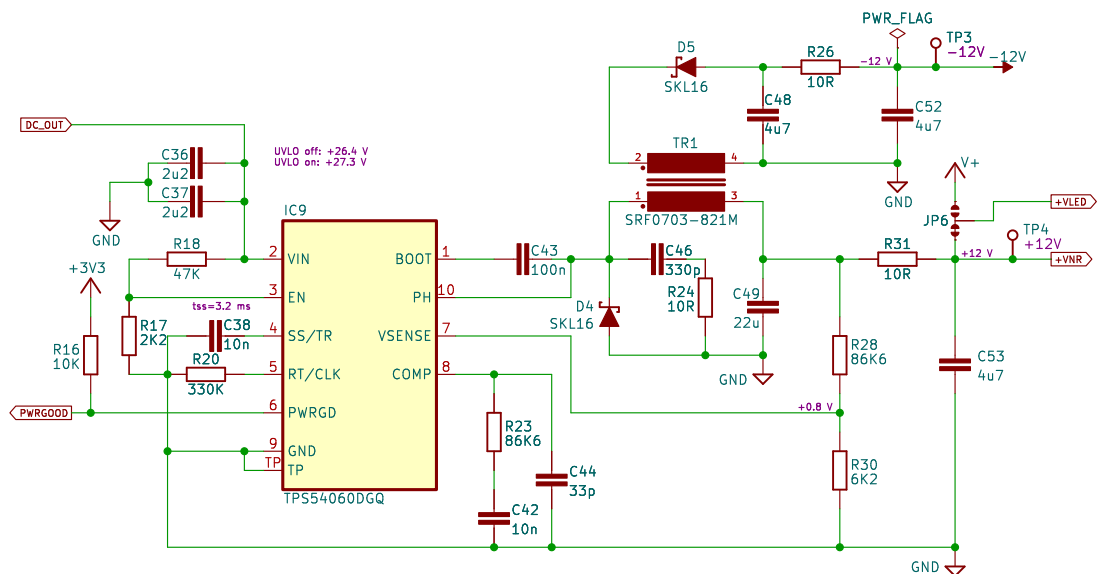
Date: 2026-02-12

Rev: r4B4

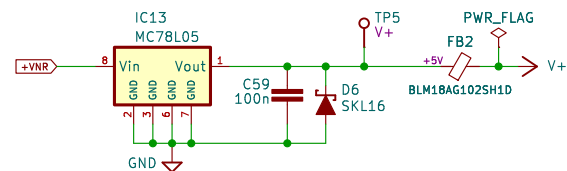
KiCad E.D.A. 9.0.7

Id: 3/8

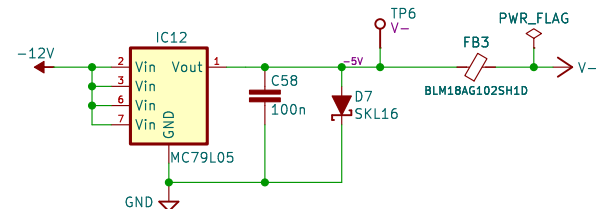
## Bias power supply



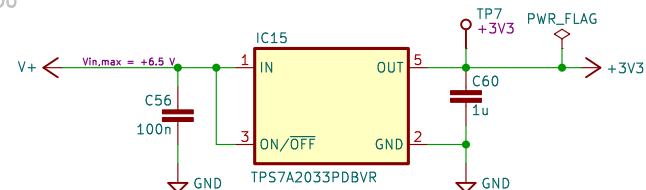
## +5V LDO



## -5V LDO



## +3V3 LDO



<https://www.envox.eu>

Power module with SMPS pre-regulator, remote sense & programming  
Licensed under the TAPR Open Hardware License ([www.tapr.org/OHL](http://www.tapr.org/OHL))

**Envox d.o.o.**

Sheet: /Bias power/

File: Bias\_power.kicad\_sch

**Title: EEZ DIB DCP405+ power module**

Size: A4

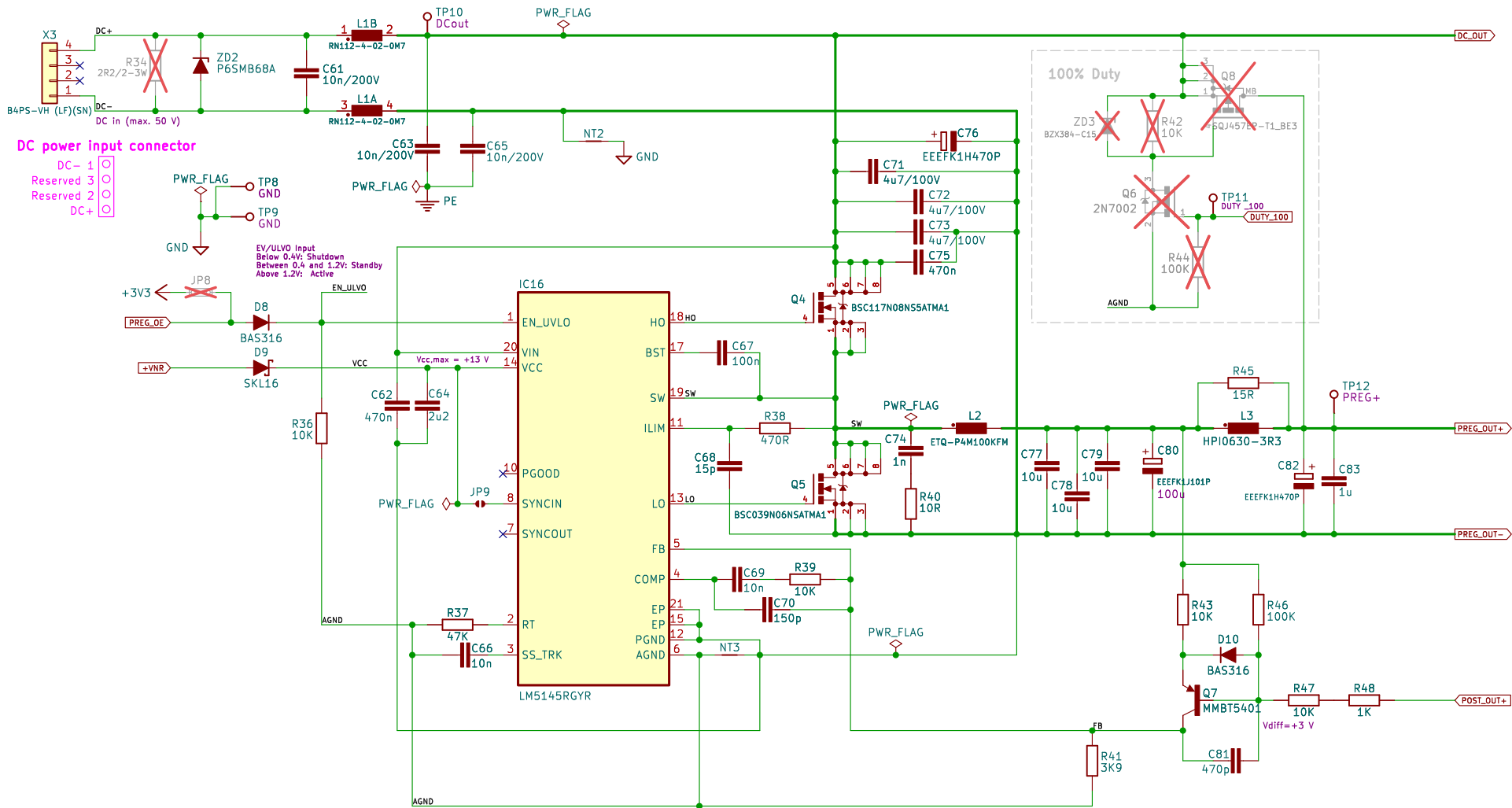
Date: 2026-02-12

Rev: r4B4

KiCad E.D.A. 9.0.7

Id: 3/8

# Low EMI power pre-regulator



<https://www.envox.eu>  
Power module with SMPS pre-regulator, remote sense & programming  
Licensed under the TAPR Open Hardware License ([www.tapr.org/OHL](http://www.tapr.org/OHL))

**Envox d.o.o.**

Sheet: /Power pre-regulator/  
File: Power\_pre-regulator.kicad\_sch

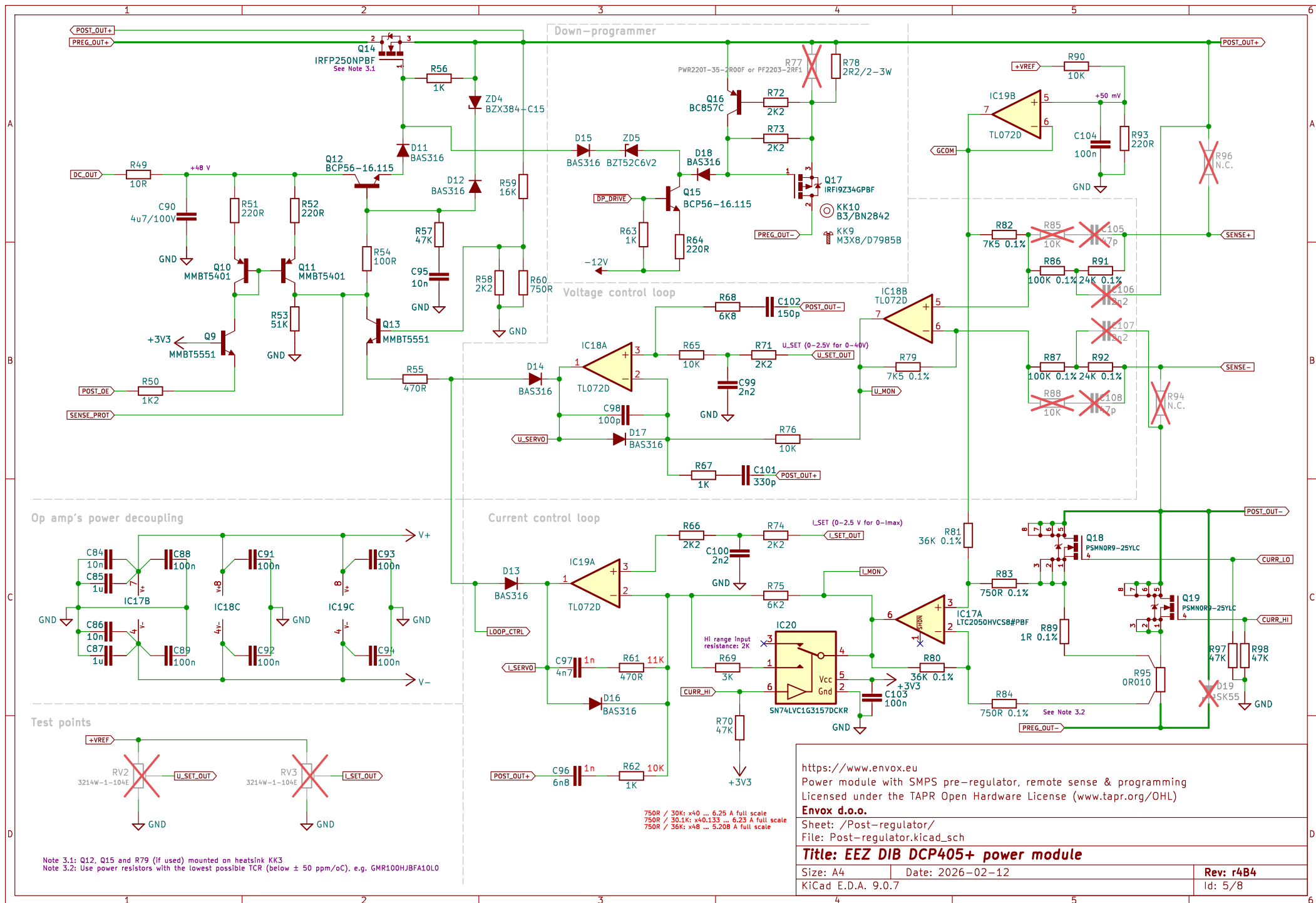
**Title: EEZ DIB DCP405+ power module**

Size: A4 Date: 2026-02-12

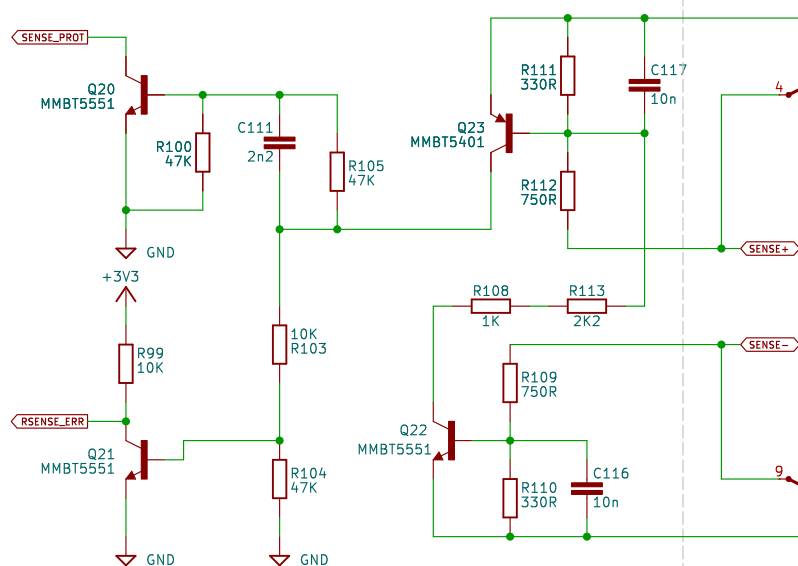
KiCad E.D.A. 9.0.7

Rev: r4B4

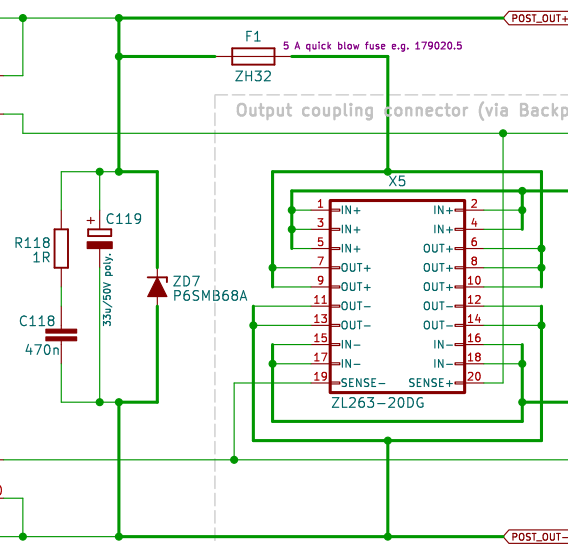
Id: 4/8



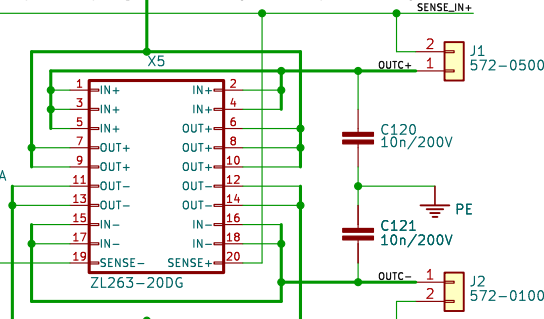
## Remote sense reverse polarity detection



## Power output section



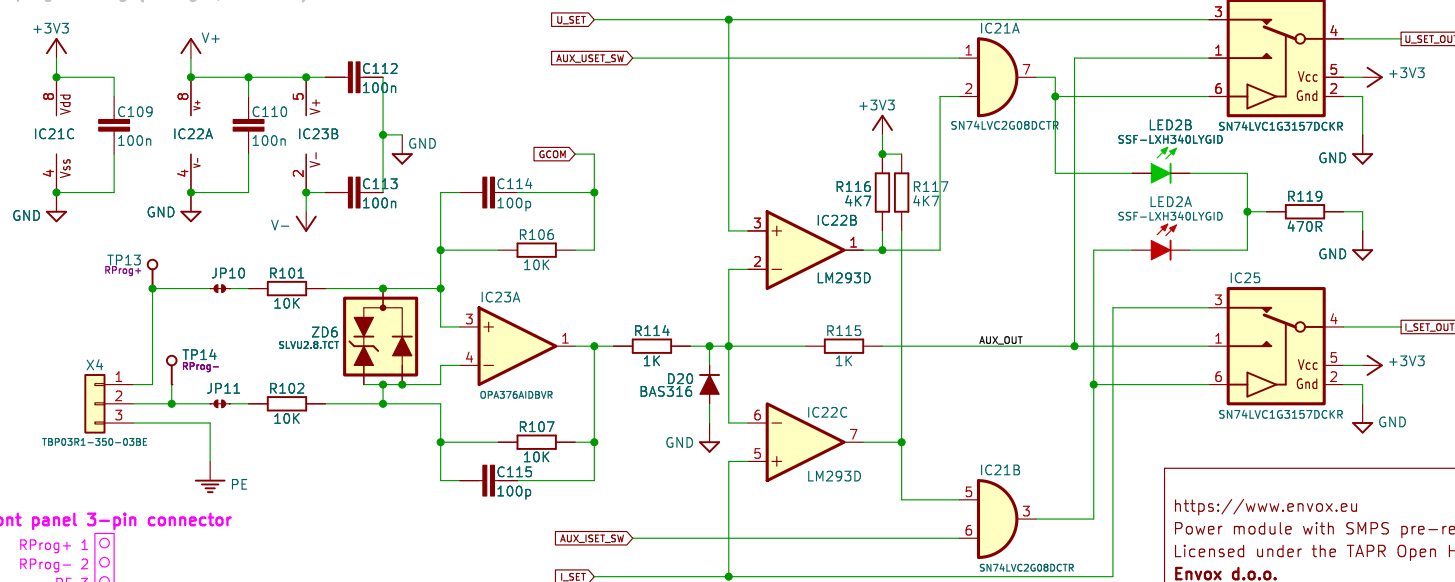
## Output coupling connector (via Backplane PCB)



## 20-pin DIB 2.0 power socket

|              |              |
|--------------|--------------|
| OUTC+ 1      | 2 OUTC+      |
| OUTC+ 3      | 4 OUTC+      |
| OUTC+ 5      | 6 POST_OUT+  |
| POST_OUT+ 7  | 8 POST_OUT+  |
| POST_OUT+ 9  | 10 POST_OUT+ |
| POST_OUT- 11 | 12 POST_OUT- |
| POST_OUT- 13 | 14 POST_OUT- |
| POST_OUT- 15 | 16 OUTC-     |
| OUTC- 17     | 18 OUTC-     |
| SENSE- 19    | 20 SENSE+    |

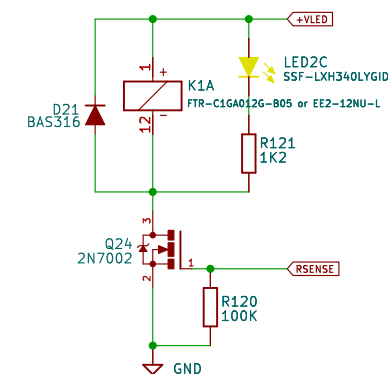
## Remote programming (Voltage / Current)



## Front panel 3-pin connector

|          |   |
|----------|---|
| RProg+ 1 | 2 |
| RProg- 2 | 3 |
| PE 3     |   |

## Remote sense selection



<https://www.envox.eu>

Power module with SMPS pre-regulator, remote sense & programming  
Licensed under the TAPR Open Hardware License ([www.tapr.org/OHL](http://www.tapr.org/OHL))

**Envox d.o.o.**

Sheet: /Rsense protection, power output, Rprog/

File: Rsense\_protection\_power\_output.kicad\_sch

**Title: EEZ DIB DCP405+ power module**

Size: A4

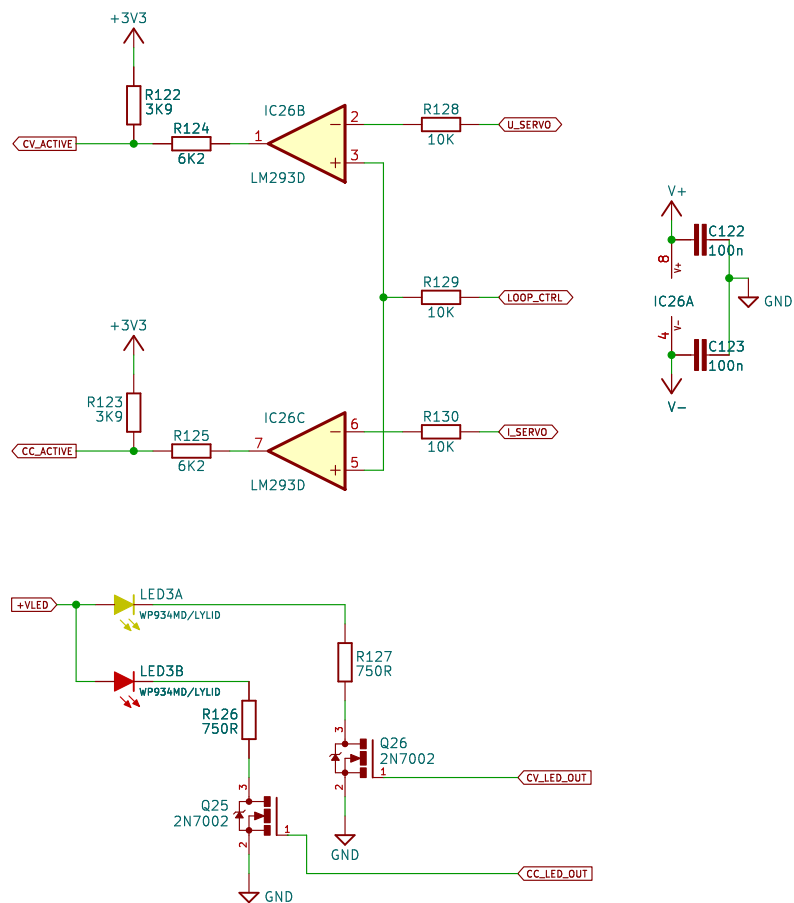
Date: 2026-02-12

KiCad E.D.A. 9.0.7

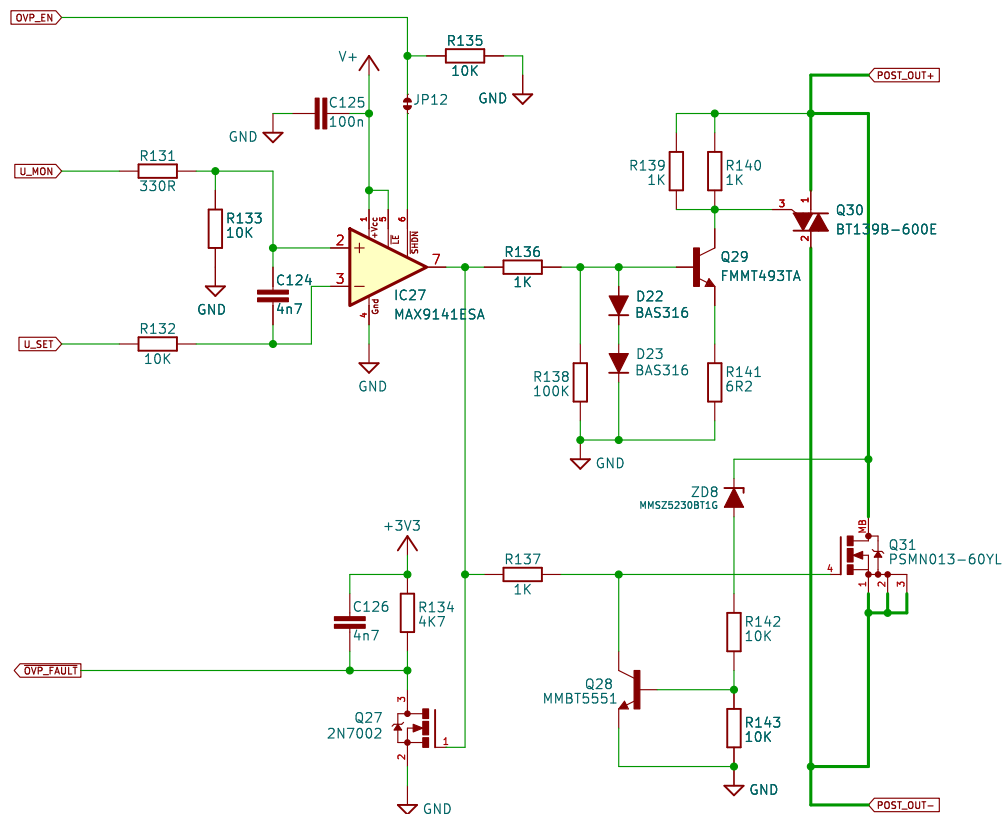
Rev: r4B4

Id: 6/8

# Constant Voltage (CV) and Constant Current (CC) mode detection and indication



# Over-voltage protection (OVP)



<https://www.envox.eu>

Power module with SMPS pre-regulator, remote sense & programming  
Licensed under the TAPR Open Hardware License ([www.tapr.org/OHL](http://www.tapr.org/OHL))

**Envox d.o.o.**

Sheet: /Mode detection, OVP/

File: CC\_CV\_OVP.kicad\_sch

**Title: EEZ DIB DCP405+ power module**

Size: A4

Date: 2026-02-12

Rev: r4B4

KiCad E.D.A. 9.0.7

Id: 7/8